Table 1: Skill routing performance. Metrics are grouped beneath each dataset. Bold/underline mark best/second distinct value within each model, retriever, dataset, and metric (ties retained). nDCG@1 and nDCG@10 are supplied source metrics.

Method	bigcodebench		champ		logicbench		medcalcbench		theoremqa		toolqa		Average	
	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10
BM25														
BM25	0.610	0.544	0.423	0.410	0.158	0.220	0.315	0.462	0.524	0.659	0.070	0.271	0.350	0.428
LLM Select	0.504	0.534	0.462	0.439	0.202	0.309	0.533	0.633	**0.695**	**0.791**	0.507	0.595	0.484	0.550
Progressive Disclosure	0.691	0.570	0.423	0.410	0.175	0.234	0.388	0.509	0.524	0.657	0.084	0.281	0.381	0.444
SkillDAG	0.610	0.589	0.423	0.442	0.158	0.315	0.309	0.461	0.524	0.663	0.070	0.303	0.349	0.462
Skill-Pro	0.610	0.543	0.423	0.410	0.158	0.221	0.309	0.458	0.524	0.659	0.070	0.271	0.349	0.427
RouterBSI	**0.878**	**0.729**	**0.500**	**0.498**	**0.281**	**0.371**	**0.818**	**0.842**	0.638	0.748	**0.577**	**0.665**	**0.615**	**0.642**
RouterBSI_init	0.431	0.447	0.308	0.367	0.158	0.281	0.642	0.720	0.448	0.627	0.256	0.413	0.374	0.476
SkillSight														
SkillSight	0.724	0.616	0.308	0.429	0.167	0.406	0.830	0.903	0.733	0.852	0.102	0.390	0.477	0.599
LLM Select	0.439	0.563	0.385	0.475	0.360	0.536	0.842	0.924	0.667	0.828	0.442	0.582	0.522	0.651
Progressive Disclosure	0.707	0.623	0.308	0.429	0.211	0.438	0.842	0.908	0.733	0.852	0.116	0.395	0.486	0.608
SkillDAG	0.724	0.668	0.308	0.449	0.167	0.455	0.830	0.905	0.733	0.849	0.102	0.398	0.477	0.621
Skill-Pro	0.724	0.614	0.308	0.429	0.167	0.406	0.830	0.903	0.733	0.852	0.107	0.392	0.478	0.599
RouterBSI	**0.862**	**0.771**	**0.577**	**0.558**	**0.439**	**0.608**	**0.933**	**0.960**	**0.752**	**0.857**	**0.702**	**0.797**	**0.711**	**0.759**
RouterBSI_init	0.585	0.465	0.462	0.510	0.211	0.449	0.879	0.927	0.648	0.804	0.316	0.505	0.517	0.610